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Robust trajectory optimization under arbitrary uncertainty using ensemble weather forecasting and statistical aircraft performance model

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ABSTRACT

Aircraft trajectory optimization can improve delay absorption and fuel efficiency; however, existing methods suffer from problems such as insufficient adaptability to uncertainty. This paper introduces a robust trajectory optimization algorithm for optimizing flight profiles in the presence of arbitrary uncertainties in weather forecasting and aircraft performance models. Ensemble forecast data are used to provide the uncertainty in the weather forecasts, and a statistical aircraft performance model based on flight data is utilized to provide the uncertainty in the aircraft performance model. Arbitrary polynomial chaos expansion is applied to efficiently quantify arbitrary uncertainties in aircraft dynamics, and incorporated into the trajectory optimization algorithm. The β-hill climbing algorithm, a stochastic local search algorithm developed recently, is employed to determine the optimal flight profiles. By conducting numerical simulations on a popular domestic flight route in Japan during time-based metering operations, this study assesses potential fuel savings by the optimal trajectories with speed control and step descent along the planned route, as opposed to radar vectoring. Actual flight data from past flights are utilized for evaluation, and average fuel savings of approximately 2% are expected. Based on the analysis conducted in this study, the effectiveness of the robust trajectory optimization algorithm and fuel savings for time-based metering operations are evaluated and demonstrated.

1. Introduction

In the past few decades, there has been a significant surge in the demand for aviation. Despite the worldwide COVID-19 pandemic, long-term growth in the aviation sector is anticipated to resume gradually (Gudmundsson et al., 2021). A fundamental concept for enhancing the safety and efficiency of air transportation while accommodating growing aviation demand is four-dimensional (4D) trajectory-based operations (TBO), which accurately describe an aircraft's path in three-dimensional space and time. TBO aims to strategically manage an aircraft's 4D trajectory, with time-based metering operations essential in its implementation. During these operations, flights receive instructions from air traffic control (ATC) to cross specific waypoints (fixes) along their routes at designated times to facilitate merging, spacing, and sequencing. If necessary, pilots adjust the aircraft speed to meet the assigned flyover time. Therefore, time-based metering operations rely on proper speed control to absorb delays.

Numerous studies have been conducted on speed control strategies (Prats and Hansen, 2011; Delgado and Prats, 2012; Xu et al., 2017). Instead of typical airborne delay through airborne holding and radar vectoring, the speed control strategy only adjusts the

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speed to absorb the assigned delay and allows the aircraft to fly its planned route. In terms of fuel consumption, typical airborne holding consumes more fuel because of path stretching (i.e., the actual trajectory is extended beyond the initially planned trajectory). The most promising feature of speed control is that airborne delay absorption can be flexibly managed along the initially planned trajectory without incurring extra fuel compared to that in the initial plan.

For tactical delay management in time-based metering operations, Matsuno and Andreeva-Mori (2020) statistically analyzed the airborne delay and potential fuel savings achievable by cruise speed control along an initially planned route without path stretching, based on data of actual past flights. Dalmau and Prats (2017) analyzed the feasible time window achieved by idle thrust descent and the associated fuel consumption for aircraft that had already initiated their descent. Additionally, a recent study by Sáez and Prats (2023) compared the fuel efficiency of idle thrust descents along renegotiated routes versus powered descents along published routes. The study found that the renegotiated route with path stretching consumed less fuel under certain conditions, such as specific wind patterns. This suggests that path stretching does not necessarily lead to increased fuel consumption.

As demonstrated in the aforementioned studies by Xu et al. (2017), Dalmau and Prats (2017), Sáez and Prats (2023), trajectory optimization represents a hopeful avenue in TBO exploration, with various algorithms developed to determine optimal flight profiles. Optimal control stands out as a powerful framework for producing optimal aircraft trajectories. A widely used method for numerically solving optimal control problems is direct methods for nonlinear programming (Dalmau et al., 2018). Direct methods effectively handle constraints in optimal control problems, and they have successfully solved complex, realistic commercial aircraft trajectory optimization problems under a tightly regulated air traffic management (ATM) procedure. The direct method (Dalmau et al., 2018) has been applied, for example, in the study by Delgado et al. (2024), which proposed a framework for in-flight trajectory optimization that accounts in advance for the predicted holding time in the terminal maneuvering area.

As another method for solving optimal control problems, Park et al. (2017) considered the vertical path for climb and descent and proposed an in-flight trajectory optimization method using a semi-analytic solution approach based on the implicit formulation of the optimal singular arc. Furthermore, Sáez et al. (2021) proposed an integrated trajectory and sequence optimization method to manage arrival traffic in terminal airspace. Lateral arrival routes are also considered to ensure separation of arrival traffic and to enable continuous descent operations (CDOs) with idle thrust. The approach by Park et al. (2017) is applied for vertical trajectory optimization, while a method based on mixed-integer programming, proposed by Dahlberg et al. (2018), Sáez et al. (2020), is used for lateral trajectory optimization.

Metaheuristics represent another widely used approach apart from optimal control. The trajectory of a commercial aircraft follows a typical flight pattern, which can be characterized as a set of static parameters consisting of speed and altitude. Metaheuristic methods are suitable for combinatorial optimization; trajectory optimization methods using genetic algorithms (Patrón and Botez, 2015), ant colony optimization (Murrieta-Mendoza et al., 2017), and artificial bee colony optimization (Murrieta-Mendoza et al., 2018) have been proposed.

However, these studies applying optimal control and metaheuristics use deterministic approaches that do not consider the uncertainties inherent in aircraft operations. As noted by Cook et al. (2015), an aircraft trajectory is affected by various uncertainties, including uncertainties in weather, data (e.g., the gap between the aircraft performance model and actual performance), and operations (e.g., pilot' intent). These uncertainties result in uncertainty in the aircraft trajectory. Deterministic approaches cannot evaluate the effects of the causing uncertainty or its propagation into the resulting uncertainty. Because insufficient consideration of uncertainty can lead to a significant gap between the planned and actual trajectories, it is vital to consider the effects of uncertainties on the aircraft trajectory.

More recently, trajectory optimization has been extended to stochastic approaches to account for uncertainty. A robust trajectory optimization approach under uncertainty typically creates and optimizes a set of trajectories, each of which involves each random element. González-Arribas et al. (2018) proposed a robust trajectory optimization for cruise flight that uses ensemble weather forecasts, handling uncertainty in weather prediction using a set of predicted weather scenarios. González-Arribas et al. (2018) dealt with wind uncertainty, while Soler et al. (2020) applied the robust trajectory optimization to convective weather uncertainty. Kamo et al. (2021) also extended this approach to descent flight. Furthermore, González-Arribas et al. (2023) proposed a fast 4D trajectory optimization that leverages general-purpose computing on graphics processing units (GPUs). Aside from ensemble weather forecasting, Matsuno et al. (2015) applied a wind prediction error model represented by a probability density function of random variables. These existing studies have proposed trajectory optimizations that incorporate uncertainty in weather prediction (ensemble forecasts or weather prediction error models), which is one of the main sources of uncertainty in aircraft operations. In addition to weather forecast uncertainty, departure time uncertainty has been incorporated into trajectory optimization (Cerezo-Magaña et al., 2022). Departure time uncertainty is also available in open data; for example, from the database[1] of the United States Bureau of Transportation Statistics, Department of Transportation, and can be utilized for trajectory optimization.

In addition, as mentioned by Cook et al. (2015), aircraft performance models are another source of uncertainty. In the aforementioned studies, the base of aircraft data (BADA) model (Nuic et al., 2010) developed by EUROCONTROL was applied. The BADA model is a deterministic aircraft performance model that represents the performance of each aircraft type and has been utilized in many research efforts in the ATM field. However, as evaluated by Harada et al. (2013), the difference in fuel consumption between BADA predictions and actual flight data is approximately 4% on average, which represents a substantial discrepancy. Even though the performance of each aircraft of the same type varies with aging and other factors, the BADA model uses a fixed parameter model specific to each aircraft type, and thus fails to fully represent actual flight performance. Therefore,

[1] https://www.transtats.bts.gov/

in our previous study (Matsuda and Matsuno, 2023), we developed an accurate aircraft performance model based on actual flight data using Bayesian methods. Bayesian inference allows aircraft performance to be statistically modeled and estimated as a joint probability distribution (a histogram of several thousand sample points), thereby quantifying the uncertainty in the model. Thus, this study introduces a robust trajectory optimization method that considers the uncertainty in the aircraft performance model in addition to the weather prediction uncertainty considered in existing studies.

As the uncertainty in the aircraft performance model is represented as a histogram of several thousand sample points, using the entire set of trajectories corresponding to each sample point, as in the robust trajectory optimization method described above, is computationally expensive and intractable in an iterative optimization process. Therefore, in this study, we apply polynomial chaos expansion (PCE), especially arbitrary polynomial chaos expansion (aPC), which can approximate the evolution of uncertainty in the aircraft trajectory without handling the entire set of trajectories. PCE was also applied by Matsuno et al. (2015) and Cerezo-Magaña et al. (2022), but with generalized polynomial chaos expansion (gPC). That approach requires the assumption of a certain probability distribution, such as Gaussian or uniform distributions for random variables, and cannot handle arbitrary probability distributions that are specified numerically as histograms or as raw datasets, such as the uncertainty in the aircraft performance model. Therefore, this study proposes a robust trajectory optimization method that incorporates aPC, which can handle arbitrary uncertainties.

This study focuses on trajectory optimization during the cruise and descent phases, assuming time-based metering operations in which ATC assigns a required time of arrival (RTA) at a metering fix during the cruise phase. The study above by Sáez and Prats (2023) demonstrated that path stretching, when combined with idle thrust descents along renegotiated, longer routes, does not necessarily lead to increased fuel consumption. However, this approach requires trajectory optimization, including consideration of the lateral path. In contrast, this study focuses on evaluating the potential fuel savings by optimizing the vertical path to the metering fix along the initially planned route, without resorting to path stretching. The major contributions of this study are as follows:

1 The proposal of a robust trajectory optimization method that incorporates aPC to account for arbitrary uncertainty in the aircraft performance model with arbitrary probability distributions in addition to the weather forecast uncertainty.
2 An evaluation of the potential benefits of fuel savings achieved by using speed control and step descent (as discussed in Section 2) instead of path stretching in time-based metering operations by comparing the optimal solution with actual past flight data.

By using the proposed robust trajectory optimization, the potential benefits of fuel savings can be obtained as probability distributions. In addition, because the aircraft performance model performs estimations based on actual flight data, the fuel consumption results of actual past flights can be directly compared with the optimal simulation results using the robust trajectory optimization. When using a standard aircraft performance model, such as the BADA model, fuel consumption based on actual past flight data must be estimated using the BADA model (Matsuno and Andreeva-Mori, 2020; Dalmau et al., 2020) instead of using fuel consumption recorded in flight data for comparison with optimal simulation results. This results in the creation of another source of uncertainty: estimation error. Thus, a unique feature of this study is that the potential benefits of fuel savings are evaluated by directly comparing the fuel consumption results of actual past flights, as recorded in flight data, with the optimal simulation results obtained by the robust trajectory optimization.

The paper is organized as follows: Section 2 covers the aircraft dynamics and the deterministic trajectory calculation method. Section 3 introduces the uncertainties in weather forecasts and the aircraft performance model, and describes the application of aPC for uncertainty quantification. Section 4 presents the trajectory optimization method. In Section 5, potential fuel savings by the optimal trajectories are analyzed through illustrative numerical simulations. Finally, Section 6 provides the conclusions.

2. Deterministic trajectory calculation

This study proposes a robust trajectory optimization method based on metaheuristics. The trajectory optimization method is described in detail in Section 4, whereas this section describes a typical flight pattern of a commercial aircraft, which can be characterized as a set of static parameters consisting of speed and altitude.

2.1. Aircraft dynamics

For typical commercial flight operations, the lateral path is defined by a series of predetermined fixes along the planned route. Hence, this study solely focuses on the vertical dynamics of the aircraft, as given by the following equations:

$$\frac{ds}{dt} = v_g \tag{1}$$

$$\frac{dh}{dt} = v \sin \gamma_a \tag{2}$$

$$\frac{dv}{dt} = \frac{Th - D}{m} - g \sin \gamma_a - \left(\frac{dw_n}{dt} \cos \chi + \frac{dw_e}{dt} \sin \chi \right) \cos \gamma_a \tag{3}$$

$$\frac{dm}{dt} = -FF \tag{4}$$

where t is the time; s is the along-track distance; v_g is the ground speed; h is the geodetic altitude; v is the true airspeed; γ_a is the air flight path angle; Th and D are the thrust and drag forces, respectively; m is the aircraft weight; g is the acceleration due to gravity;

w_n and w_e are the wind speeds in the northward and eastward directions, respectively; dw_n/dt and dw_e/dt are the windshear terms; χ is the heading angle; and FF is the fuel flow. Note that D and FF are obtained from the aircraft performance model described in Section 3.2. v_g in Eq. (1) and χ in Eq. (3) are determined from the velocity triangle:

$$v_g = \sqrt{v^2 \cos^2 \gamma_a - w_c^2} + w_a \tag{5}$$

$$\chi = \psi + \arcsin \frac{w_c}{v \cos \gamma_a} \tag{6}$$

where w_c and w_a are the cross- and along-track components of the wind vector, respectively, and ψ is the true track angle. Because ψ depends on the lateral path, it can be represented as a predefined function of s when the lateral path is predetermined.

In addition, the wind prediction data from numerical weather prediction agencies typically consist of grid point values of w_n and w_e. Consequently, w_c and w_a in Eqs. (5) and (6) are derived from w_n, w_e, and ψ:

$$w_c = w_n \sin \psi - w_e \cos \psi$$

$$w_a = w_n \cos \psi + w_e \sin \psi$$

Because we obtain w_n and w_e as grid point values, we can use them as a prescribed function of s. Accordingly, w_c and w_a are given by a prescribed function of s.

In this study, as discussed in EUROCONTROL Experimental Centre (2010), Toratani et al. (2020), we use the pressure altitude h_p instead of h, as follows:

$$\frac{dh_p}{dt} = ROCD \tag{7}$$

where $ROCD$ is the rate of climb or descent, representing the variation of h_p with time. Assuming a standard constant gravity field, the differentials of geodetic altitude and geopotential altitude are equivalent, and the ratio between the differentials of geopotential (or geodetic) altitude and pressure altitude can be expressed as (EUROCONTROL Experimental Centre, 2010):

$$\frac{dh}{dh_p} = \frac{T}{T_{\mathrm{ISA}}} \tag{8}$$

where T_{ISA} and T represent the temperatures in the International Standard Atmosphere (ISA) and the non-standard atmosphere, respectively. This equation shows that the ratio between the changes in h and h_p corresponds to the ratio between the temperature at that location and the ISA temperature at the same location. Consequently, from Eqs. (2), (7), and (8), Eq. (3) can be rewritten as the following equation:

$$\frac{dv}{dt} = \frac{Th - D}{m} - \frac{gT}{vT_{\mathrm{ISA}}} ROCD - \left(\frac{dw_n}{dt} \cos \chi + \frac{dw_e}{dt} \sin \chi \right) \cos \gamma_a \tag{9}$$

Afterward, the aircraft trajectory is determined through numerical integration of Eqs. (1), (4)–(7), and (9). The specific process is outlined in the subsequent subsection.

2.2. Trajectory calculation

Time-based metering operations are expected to be conducted during both the cruise and descent phases. The aircraft maintains a consistent Mach number and altitude during the cruise phase. Typically, for greater fuel efficiency, the aircraft tends to climb to a higher cruise altitude as its weight decreases due to fuel consumption – a practice known as step climb, commonly employed in long-haul flights. However, this study assumes to lack a step climb during the cruise phase, because we focus on short-haul flights with a maximum cruising time of approximately one hour.

In the descent phase, the aircraft descends initially at a constant descent Mach number until reaching the crossover altitude, where the calibrated airspeed (CAS) and Mach number correspond to the same true airspeed. Subsequently, the descent continues at a constant descent CAS until reaching the terminal altitude at the metering fix. In addition, the descent phase assumes CDOs, which typically allow aircraft to descend continuously at idle thrust to reduce both fuel consumption and noise (International Civil Aviation Organization, 2010). Numerous studies have evaluated the environmental benefits of CDOs (Wubben and Busink, 2000; Clarke et al., 2004). Recently, del Pozo Domínguez et al. (2023) proposed new performance indicators to assess CDO fuel efficiency more accurately.

Generally, CDOs are performed using an optimized profile descent (OPD), calculated by the onboard flight management system. However, OPD trajectories depend on aircraft performance and weather conditions, making them difficult for air traffic controllers (ATCo) to predict. Consequently, while CDOs have been introduced into actual operations at various airports, such as Schiphol Airport in the Netherlands and Kansai International Airport in Japan, they are primarily conducted during periods of low traffic density, such as late at night or early in the morning.

This study focuses on a fixed flight-path angle (fixed-FPA) descent (Wu et al., 2015), a type of CDO. In fixed-FPA descents, the vertical path is determined by specifying an FPA (which differs from the air flight path angle, as explained later), improving trajectory predictability for ATCo and operational feasibility compared to OPD (Toratani et al., 2020). Fixed-FPA descents have been successfully demonstrated at Kansai International Airport in Japan using scheduled commercial flights, and the high ATC approval rate underscores the feasibility and acceptance of this approach (Iwata et al., 2024). Although the optimal descent FPA for fuel efficiency varies with aircraft performance and weather conditions, selecting the appropriate FPA can achieve fuel savings

comparable to those of OPD (Iwata et al., 2024). While this study focuses on fuel efficiency, fixed-FPA descents also offer the potential for noise reduction through flexible designs, such as a steeper angle approach and displaced landing (Filippone, 2017).

Aircraft adjust their flyover times at a metering fix for time-based metering operations. Speed control is typically used to adjust the flyover time without stretching the lateral path. Thus, current ATM operations increase the airborne delay by extending the lateral path through radar vectoring, whereas time-based metering operations absorb the assigned delay only through speed control on the planned lateral path. However, De Smedt et al. (2013) and Matsuno and Andreeva-Mori (2022) analyzed the feasibility of time-based metering operations during the descent phase, and mentioned that speed control alone is sometimes insufficient to absorb the assigned delay. In such cases, a combination of speed control and a step descent to a lower altitude may be necessary to increase airborne delay. Generally, more delay can be achieved by descending to a lower cruise altitude, where the aircraft may fly at a reduced true airspeed. However, this effect depends on the wind conditions: ground speed at lower altitudes might increase if stronger tailwinds are encountered, which could reduce the delay-absorbing potential. Despite these potential variations, both studies (De Smedt et al., 2013; Matsuno and Andreeva-Mori, 2022) demonstrated that speed control, when combined with a step descent, generally increases airborne delay. Therefore, this study considers a step descent when the assigned delay cannot be absorbed by speed control alone during a continuous descent.

Next, we describe the trajectory calculation. s in Eq. (1) consistently increases relative to t, because v_g in Eq. (5) remains positive for the fixed-wing aircraft considered in this study. In addition, during the descent phase, h_p in Eq. (7) monotonically decreases concerning t. Consequently, instead of employing t as the independent variable, we utilize s and h_p for the cruise and descent phases, respectively.

In the cruise phase, Eqs. (1), (4)–(7), and (9) can be reformulated as follows:

$$\frac{dt}{ds} = \frac{1}{v_g} \tag{10}$$

$$\begin{aligned}\frac{dv}{ds} &= \frac{dv}{dt}\frac{dt}{ds} \\ &= \frac{Th - D}{mv_g} - \left(\frac{dw_n}{ds}\cos\chi + \frac{dw_e}{ds}\sin\chi\right)\end{aligned} \tag{11}$$

$$\begin{aligned}\frac{dm}{ds} &= \frac{dm}{dt}\frac{dt}{ds} \\ &= -\frac{FF}{v_g}\end{aligned} \tag{12}$$

$$v_g = \sqrt{v^2 - w_c^2} + w_a \tag{13}$$

$$\chi = \psi + \arcsin\frac{w_c}{v} \tag{14}$$

where $ROCD = 0$ and γ_a is sufficiently small to approximate $\cos\gamma_a \simeq 1$ during the cruise phase. As described in Section 3.2, D in Eq. (11) can be calculated using the cruise Mach number. Then, we can calculate Th from Eq. (11) under the condition that the cruise Mach number remains constant during the cruise phase. Thus, we can calculate the trajectory during the cruise phase by numerically integrating Eqs. (10)–(12).

In the descent phase, the following equation is obtained from Eqs. (7) and (9):

$$\frac{dt}{dh_p} = \frac{m}{Th - D}\left(\frac{dv}{dh_p} + \frac{gT}{vT_{\mathrm{ISA}}} + \frac{dw_n}{dh_p}\cos\chi + \frac{dw_e}{dh_p}\sin\chi\right) \tag{15}$$

In commercial air transport operations, γ_a is sufficiently small enough to approximate $\cos\gamma_a \simeq 1$ in Eq. (15). Consequently, v_g and χ are assumed to be equivalent to those in Eqs. (13) and (14), respectively. Eqs. (1) and (4) can be rewritten as follows:

$$\begin{aligned}\frac{ds}{dh_p} &= \frac{ds}{dt}\frac{dt}{dh_p} \\ &= \frac{mv_g}{Th - D}\left(\frac{dv}{dh_p} + \frac{gT}{vT_{\mathrm{ISA}}} + \frac{dw_n}{dh_p}\cos\chi + \frac{dw_e}{dh_p}\sin\chi\right)\end{aligned} \tag{16}$$

$$\begin{aligned}\frac{dm}{dh_p} &= \frac{dm}{dt}\frac{dt}{dh_p} \\ &= -\frac{mFF}{Th - D}\left(\frac{dv}{dh_p} + \frac{gT}{vT_{\mathrm{ISA}}} + \frac{dw_n}{dh_p}\cos\chi + \frac{dw_e}{dh_p}\sin\chi\right)\end{aligned} \tag{17}$$

As either the CAS or Mach number remains constant during the descent phase, dv/dh_p can be computed when either value is given. For a fixed-FPA descent, the following equation holds:

$$\frac{dh_p}{ds} = \tan\gamma_{\mathrm{fix}} \tag{18}$$

where γ_{fix} is the fixed FPA. Note that the FPA in fixed-FPA descents is different from air flight path angle γ_a. Then, from Eqs. (16) and (18), Th is determined using the following equation:

$$Th = D + mv_g\tan\gamma_{\mathrm{fix}}\left(\frac{dv}{dh_p} + \frac{gT}{vT_{\mathrm{ISA}}} + \frac{dw_n}{dh_p}\cos\chi + \frac{dw_e}{dh_p}\sin\chi\right)$$

Furthermore, Eqs. (15)–(17) can be rewritten as follows:

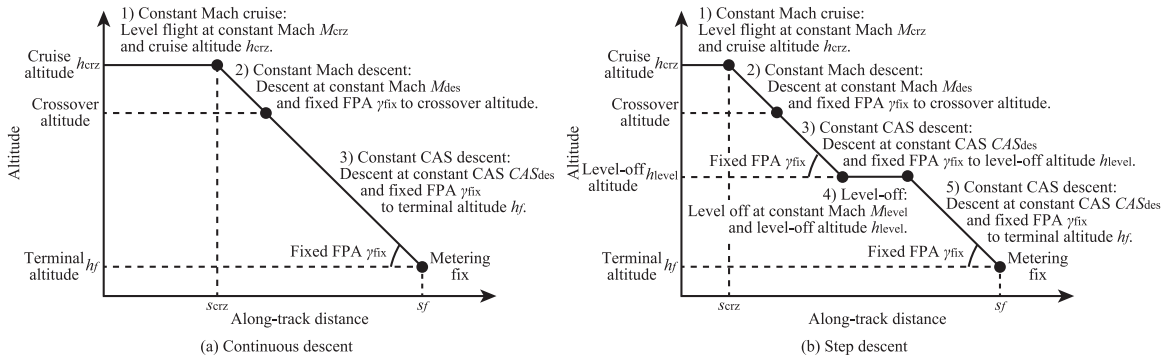

Fig. 1. Schematic descriptions of flight phases for continuous and step descent.

$$\frac{dt}{dh_p} = \frac{1}{v_g \tan\gamma_{\text{fix}}} \tag{19}$$

$$\frac{ds}{dh_p} = \frac{1}{\tan\gamma_{\text{fix}}} \tag{20}$$

$$\frac{dm}{dh_p} = -\frac{FF}{v_g \tan\gamma_{\text{fix}}} \tag{21}$$

Thus, we can calculate the trajectory during the descent phase by numerically integrating Eqs. (19)–(21).

Fig. 1 illustrates the flight phases and descriptions used in this study. Note that the level-off phase during step descent may be in a different order than in Fig. 1, depending on the combination of level-off and crossover altitudes. Here, the along-track distance to the metering fix s_f is known because the lateral path is predefined. Thus, the cruise distance s_{crz} in a continuous descent is obtained as $s_f - (h_{\text{crz}} - h_f)\tan\gamma_{\text{fix}}$, where h_{crz} and h_f are the cruise and terminal altitudes, respectively. In contrast, the cruise distance s_{crz} in a step descent can be expressed as $r_{\text{crz}}(s_f - (h_{\text{crz}} - h_f)\tan\gamma_{\text{fix}})$ by introducing the ratio of the cruise distance to the sum of the cruise distance and the level-off distance r_{crz}. Therefore, as described above, the flight profile for a continuous descent is calculated using four variables: the cruise Mach number M_{crz}, fixed FPA γ_{fix}, descent Mach number M_{des}, and descent CAS CAS_{des}. The flight profile for a step descent is calculated using seven variables: the cruise Mach number M_{crz}, ratio of cruise distance r_{crz}, fixed FPA γ_{fix}, descent Mach number M_{des}, descent CAS CAS_{des}, Mach number during level-off M_{level}, and level-off altitude h_{level}. In this study, both CAS_{des} and γ_{fix} are assumed to remain constant before and after level-off to avoid adding additional decision variables. Note that h_{crz} and h_f are derived from actual flight data. These decision variables are optimized using trajectory optimization as described in Section 4.

3. Uncertainty quantification

In this section, we introduce the uncertainties in the weather prediction and in the aircraft performance model, and describe aPC for uncertainty quantification.

3.1. Ensemble weather forecasting

Weather predictions are crucial for ATM research because weather significantly influences the efficiency of aircraft trajectory. Despite advancements in weather forecasting techniques over recent decades, they are not infallible, and uncertainties remain even in cutting-edge numerical weather prediction (NWP) systems. The uncertainties in NWP primarily stem from imprecise atmospheric models, errors in observations, and imprecise and incomplete knowledge of atmospheric conditions. These factors contribute to the chaotic nature of the atmosphere, where even minor errors in a deterministic NWP model can rapidly increase over time, leading to predictions diverging significantly from actual weather conditions. To address uncertain weather forecasts, ensemble weather forecasting has been developed to provide probabilistic forecasts and estimate weather prediction uncertainty (World Meteorological Organization, 2012; Bauer et al., 2015). Ensemble weather forecasting entails running multiple simulations, known as "ensemble members", with perturbations in the initial conditions or atmospheric parameters. These forecasts encompass a wide range of potential weather outcomes, including the actual weather. Operational NWP centers worldwide maintain ensemble prediction systems that employ various ensemble forecasting techniques to capture uncertainties with a limited number of ensemble members, typically ranging from 10 to 50. Through this approach, probabilistic measures such as probability density function, ensemble average (mean among ensemble members), ensemble spread (standard deviation among ensemble members), and probability of an event (proportion of members in which a certain event is forecast to occur) can be derived. In ATM research, ensemble forecasts have recently been used for trajectory prediction (Franco et al., 2018) and trajectory planning (González-Arribas et al., 2018). Because weather uncertainty varies hourly, employing ensemble forecasts capable of estimating hour-to-hour uncertainty is crucial.

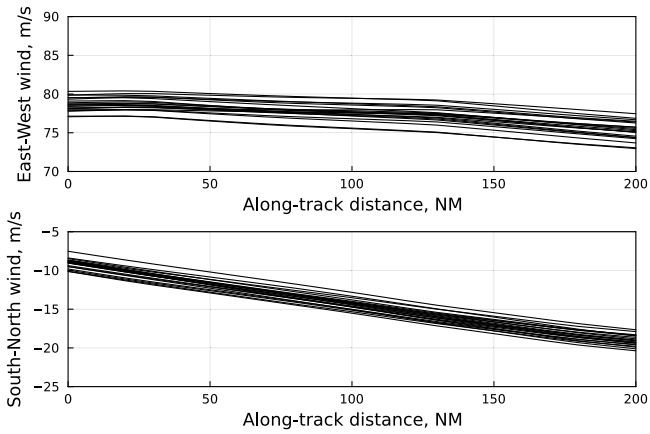

Fig. 2. Predicted winds for 21 ensemble members along flight path.

This study employs the ensemble forecast dataset provided by the Global Ensemble Forecast System (GEFS) of the National Oceanic and Atmospheric Administration.[2] The GEFS data include atmospheric properties such as wind and temperature on a three-dimensional grid. Three-dimensional grid points are placed every degree in longitude and latitude at pressure altitudes with 50 hPa intervals. In the trajectory calculation, wind and temperature data (w_n, w_e, and T) are spatially interpolated to match the aircraft trajectory using linear interpolation. Additionally, the wind derivatives (dw_n/ds, dw_e/ds, dw_n/dh_p, and dw_e/dh_p) are obtained using a first-order finite difference at a given location, as employed by Ghazi et al. (2021). There are four runs per day (0000, 0600, 1200, and 1800 Coordinated Universal Time (UTC)), and the number of ensemble members N_{ens} in each run is 21. Fig. 2 illustrates the predicted winds for 21 ensemble members along a flight path at a given cruise altitude. As detailed in Section 2, the trajectory (s, h_p, v, m) can be calculated using Eqs. (1), (4)–(7), and (9) for every member of the ensemble forecast dataset. We can then obtain N_{ens} different trajectory calculation results and use them to calculate the expected value of the trajectory under weather prediction uncertainty.

3.2. Statistical aircraft performance model

This subsection briefly reviews our previously developed statistical aircraft performance model based on flight data obtained from an on-board flight data recorder (Matsuda and Matsuno, 2023). The drag force D and fuel flow FF required for trajectory calculation are derived using the statistical model of aircraft performance. First, D is given by

$$D = \frac{1}{2}\rho v^2 S C_D$$
$$C_D = C_{D,0} + C_{D,L}C_L^2 + C_{D,M}\max(0, M - C_M)^4 + C_{D,\mu}\mu \tag{22}$$

where S is the wing area; ρ and μ are the air density and viscosity, respectively; $C_{D,0}$, $C_{D,L}$, $C_{D,M}$, $C_{D,\mu}$, and C_M are parameters to be estimated from flight data; M is the Mach number corresponding to the true airspeed v; and C_L is the lift coefficient, calculated as $2mg/\rho v^2 S$, assuming that the flight path angle γ_a is sufficiently small and that the aircraft weight m and lift force are balanced. C_D in Eq. (22) is based on the quadratic drag polar assumption, and the third and fourth terms on the right-hand side of Eq. (22) represent the compressibility and viscosity effects, respectively. In addition, FF is calculated as follows:

$$FF = \max(FF_{\text{idle}}, FF_{\text{nominal}})$$
$$FF_{\text{idle}} = C_{\zeta,0} + C_{\zeta,M}M + C_{\zeta,h_p}h_p \tag{23}$$
$$FF_{\text{nominal}} = (1 - I_{\text{crz}}C_{\eta,\text{crz}})(C_{\eta,0}\theta^{C_{\eta,\theta}}(1+M)^{C_{\eta,M}})Th \tag{24}$$

where I_{crz} is the cruise phase indicator, which is equal to 1 for the cruise phase and 0 otherwise; θ is the temperature ratio, expressed as the ratio of the temperature T at the pressure altitude h_p to the temperature at mean sea level; and $C_{\zeta,0}$, $C_{\zeta,M}$, C_{ζ,h_p}, $C_{\eta,\text{crz}}$, $C_{\eta,0}$, $C_{\eta,\theta}$, and $C_{\eta,M}$ are parameters to be estimated from flight data. Notably, two types of fuel flow are considered, FF_{idle} and FF_{nominal}, and the greater of the two is selected as the fuel flow at any given time. FF_{idle} in Eq. (23) is the minimum fuel flow required to keep the engine running at idle thrust settings, and is given as a function of M and h_p in this model. In contrast, for non-idle thrust settings, FF_{nominal} in Eq. (24) is proportional to the thrust force Th and is given as a function of θ and M in this model. Moreover, Eq. (24) indicates that the fuel flow in the cruise phase can differ from that in other phases.

The statistical model of aircraft performance comprises 12 uncertain parameters,

[2] https://www.ncei.noaa.gov/products/weather-climate-models/global-ensemble-forecast

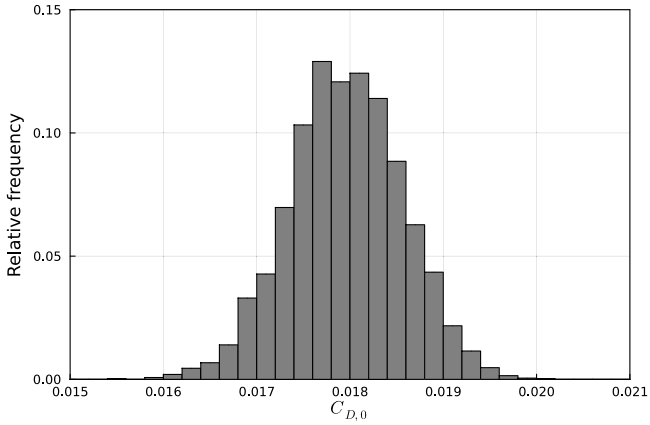

Fig. 3. Histogram for one parameter in statistical aircraft performance model.

$\boldsymbol{X} = (C_{D,0}, C_{D,L}, C_{D,M}, C_{D,\mu}, C_M, C_{\zeta,0}, C_{\zeta,M}, C_{\zeta,h_p}, C_{\eta,\mathrm{crz}}, C_{\eta,0}, C_{\eta,\theta}, C_{\eta,M})^\top$,

which are estimated from flight data. In our previous study (Matsuda and Matsuno, 2023), statistical inference was performed using the Bayesian approach, which assigns probability distributions to model parameters. The Hamiltonian Monte Carlo (HMC) method, a Markov chain Monte Carlo method (Gelman et al., 2015), was applied to estimate the uncertain parameters $\boldsymbol{X}$ as arbitrary probability distributions. See Matsuda and Matsuno (2023) for further details.

As an example of the estimation results, Fig. 3 shows the histogram of $C_{D,0}$, which is one of the twelve parameters in the statistical model of aircraft performance. As indicated in Fig. 3, the parameters of the statistical aircraft performance model are inferred as arbitrary probability distributions comprising sample points, and the number of sample points N_{apm} is 4,000. As with the ensemble forecast data, the trajectory (s, h_p, v, m) can be computed using Eqs. (1), (4)–(7), and (9) for every sample point in the statistical aircraft performance model $\boldsymbol{X}^{(i)}$ $(i = 1, 2, \ldots, N_{\mathrm{apm}})$. We can obtain N_{apm} different trajectory calculation results and use them to calculate the expected value of the trajectory under uncertainty in the aircraft performance model. However, it is computationally burdensome to calculate the expected trajectory values considering $N_{\mathrm{apm}}(= 4,000)$ trajectories ($N_{\mathrm{ens}} \times N_{\mathrm{apm}}(= 21 \times 4,000 = 84,000)$ trajectories when combined with the ensemble forecast data). Substantial computational resources are required, especially when iterative calculations are performed during trajectory optimization. Therefore, in this study, we applied PCE, as explained in the subsequent subsection, to compute the expected value with much fewer sample points.

3.3. Arbitrary polynomial chaos expansion

PCE is a technique for representing a random variable as a polynomial function of other random variables. It can determine the uncertainty propagation in dynamic systems with uncertain parameters. Initially introduced by Wiener (1938), homogeneous chaos, also referred to as Hermite chaos, utilized Hermite polynomials to approximate Gaussian processes. Hermite chaos is useful for analyzing stochastic processes with Gaussian random variables, and has been extensively utilized in quantifying uncertainties in physical applications. Notably, Ghanem and Spanos (1991) pioneered the combination of Hermite chaos and finite element methods for applications in solid mechanics. The gPC method further extends the capabilities of Hermite chaos for analyzing stochastic processes with various types of random variables, not just Gaussian random variables, using the corresponding orthogonal polynomials (Xiu and Karniadakis, 2002). For example, Hermite polynomials are used with Gaussian random variables and Legendre polynomials are used with uniform random variables. gPC has been widely used in engineering and applied sciences because it can efficiently deal with probabilistic uncertainty in the parameters of a dynamical system. Moreover, gPC has been utilized in several existing studies in the ATM field and has been applied to quantify wind uncertainty (Matsuno et al., 2015) and departure time uncertainty (Cerezo-Magaña et al., 2022).

gPC assumes exact knowledge of the involved probability density functions. Unfortunately, information on data distributions is limited in practical engineering applications. The statistical distribution of the model parameters can be nontrivial; for example, it may be skewed, multi-modal, or discontinuous. Depending on the modeling approach, statistical information on model parameters may be either continuous or discrete; they could exist analytically as probability density functions or numerically as histograms. Therefore, to overcome the shortcomings of gPC, the aPC method was developed for handling arbitrary distributions with arbitrary probability measures, which can be numerically quantified, such as through a histogram derived from raw data (Oladyshkin and Nowak, 2012). aPC further extends gPC by employing a global polynomial basis for arbitrary data distributions. The most important property of aPC is that it can handle probability measures defined only by their moments without additional information, and moments can be easily calculated using raw data. This study employs the aPC method for handling arbitrary distributions of uncertain parameters within the stochastic model of aircraft performance inferred using the HMC method. As shown in Section 3.2, the uncertain parameters are given numerically as histograms estimated from the flight data, making it difficult to apply gPC.

Using the PCE approach, the stochastic model response can be represented as the sum of the orthogonal polynomials of the independent random variables, as described by the subsequent equation:

$$Y(\boldsymbol{\xi}) = \sum_{m=1}^{\infty} C_m \boldsymbol{\Phi}_m(\boldsymbol{\xi}) \tag{25}$$

where $Y(\boldsymbol{\xi})$ is the stochastic model response, $\boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_{N_\xi})^\top$ denotes the independent random variables, C_m stands for the unknown expansion coefficient to be determined, $\boldsymbol{\Phi}_m(\boldsymbol{\xi})$ denotes the multivariate orthogonal polynomial basis function, and N_ξ represents the number of random variables. $\boldsymbol{\Phi}_m(\boldsymbol{\xi})$ is derived from the l_ith order univariate polynomial basis function $\phi^{(l_i)}(\xi_i)$ of each random variable ξ_i $(i = 1, 2, \dots, N_\xi)$ using the tensor product rule:

$$\boldsymbol{\Phi}_m(\boldsymbol{\xi}) = \prod_{i=1}^{N_\xi} \phi^{(l_i)}(\xi_i) \tag{26}$$

Note that a combination of l_i, $(l_1, \dots, l_{N_\xi})$, is unique to each subscript m. For computational purposes, infinite series in Eq. (25) is truncated to keep a finite number of terms:

$$Y_d(\boldsymbol{\xi}) = \sum_{m=1}^{N_{\mathrm{PCE}}} C_m \boldsymbol{\Phi}_m(\boldsymbol{\xi}) \tag{27}$$

where $Y_d(\boldsymbol{\xi})$ is the dth order PCE approximation of $Y(\boldsymbol{\xi})$, and N_{PCE} is the total number of polynomial basis functions or expansion coefficients. The dth order PCE approximation indicates that the maximum total degree of the polynomials is d. In the truncation scheme, the total degree of polynomials is not larger than d, and l_i in Eq. (26) satisfies the following condition:

$$d_m = \sum_{i=1}^{N_\xi} l_i \le d$$

where d_m $(m = 1, \dots, N_{\mathrm{PCE}})$ is the sum of the order of the univariate polynomial of the ith random variable $\phi^{(l_i)}(\xi_i)$ in Eq. (26). In the truncation strategy, because the total degree of the polynomial basis function d_m $(m = 1, \dots, N_{\mathrm{PCE}})$ is not larger than d, the total number of polynomial basis functions N_{PCE} is determined by the binomial coefficient $N_{\mathrm{PCE}} = \binom{N_\xi + d}{N_\xi}$. Note that when $m = 1$, $\boldsymbol{\Phi}_1(\boldsymbol{\xi})$ is set to the zero-order polynomial basis ($d_1 = 0$) that takes the value 1.

3.3.1. Construction of polynomials from statistical moments

The PCE method uses normalized orthogonal (orthonormal) polynomials for the univariate polynomial basis function, satisfying the subsequent condition of orthonormality:

$$E[\phi^{(k)}(\xi)\phi^{(l)}(\xi)] = \int \phi^{(k)}(\xi)\phi^{(l)}(\xi) d\Gamma(\xi) = \delta_{kl} \tag{28}$$

where $E[\cdot]$ denotes the expectation operator, δ_{kl} is the Kronecker delta function that is 1 if $k = l$ and 0 otherwise, and $\Gamma(\xi)$ is the probability measure. In the gPC method (Xiu and Karniadakis, 2002), selecting the most appropriate orthonormal polynomials relies on the characteristics of $\Gamma(\xi)$ for enhancing convergence. For example, Hermite and Legendre polynomials are employed with Gaussian and uniform random variables, respectively. (See Xiu and Karniadakis (2002) for a more detailed discussion.) The aPC method (Oladyshkin and Nowak, 2012), on the other hand, determines the orthonormal polynomials for an arbitrary $\Gamma(\xi)$, defined only by its statistical moments of ξ.

In the aPC method, the kth order polynomial basis function $\phi^{(k)}(\xi)$ $(0 \le k \le d)$ is defined by

$$\phi^{(k)}(\xi) = \sum_{i=0}^{k} p_i^{(k)} \xi^i \tag{29}$$

where $p_i^{(k)}$ denotes coefficients in $\phi^{(k)}(\xi)$. Note that when $k = 0$, the zero-order polynomial basis ($\phi^{(0)}(\xi)$) is 1. The orthonormality conditions in Eq. (28) for any kth order polynomial $\phi^{(k)}(\xi)$ with all lower-order polynomials can be written in the following forms, without loss of generality:

$$\begin{aligned} \int \sum_{i=0}^{k} p_i^{(k)} \xi^i d\Gamma(\xi) &= 0 \\ \int \sum_{i=0}^{k} p_i^{(k)} \xi^{i+1} d\Gamma(\xi) &= 0 \\ &\vdots \\ \int \sum_{i=0}^{k} p_i^{(k)} \xi^{i+k-1} d\Gamma(\xi) &= 0 \\ p_k^{(k)} &= 1 \end{aligned} \tag{30}$$

The ith raw moment of ξ is defined by

$$\mu_i = \int \xi^i d\Gamma(\xi) \tag{31}$$

where μ_i is the ith raw moment. Similarly, when ξ is presented only as a set of N_{apm} samples, $\{\xi^{(1)}, \xi^{(2)}, \dots, \xi^{(N_{\text{apm}})}\}$, μ_i can be approximated by

$$\mu_i = \frac{1}{N_{\text{apm}}} \sum_{m=1}^{N_{\text{apm}}} \xi^{(m)i}$$

With Eq. (31), Eq. (30) can be rewritten as

$$\begin{aligned} \sum_{i=0}^{k} p_i^{(k)} \mu_i &= 0 \\ \sum_{i=0}^{k} p_i^{(k)} \mu_{i+1} &= 0 \\ &\vdots \\ \sum_{i=0}^{k} p_i^{(k)} \mu_{i+k-1} &= 0 \\ p_k^{(k)} &= 1 \end{aligned}$$

Alternatively, it can be written in the following matrix form:

$$\begin{bmatrix} \mu_0 & \mu_1 & \cdots & \mu_k \\ \mu_1 & \mu_2 & \cdots & \mu_{k+1} \\ \vdots & \vdots & \ddots & \vdots \\ \mu_{k-1} & \mu_k & \cdots & \mu_{2k-1} \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} p_0^{(k)} \\ p_1^{(k)} \\ \vdots \\ p_{k-1}^{(k)} \\ p_k^{(k)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{32}$$

Therefore, the polynomial coefficients $p_i^{(k)}$ can be obtained from Eq. (32), and the univariate polynomial basis function in Eq. (29) can be constructed. Furthermore, the multivariate polynomial basis function in Eq. (26) can be obtained from Eq. (29) using the tensor product rule. As described above, in the aPC method, statistical moments are the only information regarding the input distribution required to construct the polynomial basis function. (See Oladyshkin and Nowak (2012) for a more detailed discussion.)

3.3.2. Probabilistic collocation method

Estimating the unknown expansion coefficient C_m in Eq. (27) can be achieved through either an intrusive or a non-intrusive approach. Intrusive approaches (Ghanem and Spanos, 1991; Xiu and Karniadakis, 2002) can be cumbersome and challenging to implement, especially for complex nonlinear systems. Conversely, non-intrusive approaches offer a more straightforward solution for handling general nonlinear problems. This approach involves utilizing sample points of random input variables and repeatedly running deterministic simulations. Therefore, this study employs a non-intrusive method to determine C_m. Although there are various non-intrusive approaches, such as the stochastic collocation method (Xiu and Hesthaven, 2005) and probabilistic collocation method (Li and Zhang, 2007), this study employs the probabilistic collocation method (Li and Zhang, 2007), as used in Oladyshkin and Nowak (2012). When dealing with a limited number of random input variables, the stochastic collocation method proves efficient by employing carefully chosen sample points, or collocation points (Xiu and Hesthaven, 2005). Nevertheless, as the number of random variables increases, the stochastic collocation method may encounter the challenge known as the *curse of dimensionality*. In contrast, the probabilistic collocation method is especially computationally efficient for a large number of random input variables, because the number of collocation points can be the same as the number of unknown expansion coefficients.

The selection of collocation points at a particular polynomial order mirrors the process in the Gaussian quadrature rule, where these points are derived from the roots of the succeeding higher-order orthogonal polynomial for each random variable. When the polynomial order is d, the total number of available collocation points is $(d+1)^{N_\xi}$, consistently exceeding the required number of collocation points N_{PCE}. Thus, N_{PCE} collocation points are chosen from the $(d+1)^{N_\xi}$ available collocation points. As in the Gaussian quadrature rule, the collocation points have associated weights (probabilities) and the collocation points are ranked in descending order of weights. The N_{PCE} collocation points are then selected in descending order of weights.

Using the probabilistic collocation method, C_m can be obtained by solving the following linear system:

$$\begin{bmatrix} \Phi_1(\xi^{(1)}) & \Phi_2(\xi^{(1)}) & \cdots & \Phi_{N_{\text{PCE}}}(\xi^{(1)}) \\ \Phi_1(\xi^{(2)}) & \Phi_2(\xi^{(2)}) & \cdots & \Phi_{N_{\text{PCE}}}(\xi^{(2)}) \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_1(\xi^{(N_{\text{PCE}})}) & \Phi_2(\xi^{(N_{\text{PCE}})}) & \cdots & \Phi_{N_{\text{PCE}}}(\xi^{(N_{\text{PCE}})}) \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_{N_{\text{PCE}}} \end{bmatrix} = \begin{bmatrix} Y(\xi^{(1)}) \\ Y(\xi^{(2)}) \\ \vdots \\ Y(\xi^{(N_{\text{PCE}})}) \end{bmatrix} \tag{33}$$

where $Y(\xi^{(m)})$ represents the deterministic model response at the mth collocation point $\xi^{(m)}$ $(m = 1, 2, \dots, N_{\text{PCE}})$. Therefore, Eqs. (27) and (33) determine the approximate stochastic model response $Y_d(\xi)$, representing the orthonormal polynomials of the random variables ξ.

3.3.3. Post-processing

After calculating the expansion coefficient C_m, the approximate stochastic model response $Y_d(\xi)$ can be derived from Eqs. (27) and (33). $Y_d(\xi)$ represents the distribution function of the random variables ξ and can be assessed for any given random inputs. $Y_d(\xi)$ serves as a surrogate model, also referred to as a response surface model or metamodel. Thus, an output $Y_d(\xi)$ can be predicted from a specific input ξ without conducting the original simulations. In addition, we can estimate the probability density function of $Y_d(\xi)$, despite the actual function being unknown. As $Y_d(\xi)$ is expressed through the polynomials of ξ in Eq. (27), the probability distribution of $Y_d(\xi)$ can be calculated using a Monte Carlo simulation of the random input variables ξ. Furthermore, we can analytically derive the statistical moments of $Y(\xi)$ from C_m. For example, the expected value of $Y(\xi)$ is determined by the subsequent equation:

$$\begin{aligned} E[Y(\xi)] &= \int \left[\sum_{m=1}^{\infty} C_m \Phi_m(\xi) \right] d\Gamma(\xi) \\ &= C_1 \end{aligned}$$

Note that $\Phi_1(\xi) = 1$ since $\Phi_1(\xi)$ is defined as the zero-order polynomial basis. Similarly, higher-order moments can be obtained using the expansion coefficient C_m.

Therefore, by applying the aPC method, arbitrary statistical quantities, such as various statistical moments and probability density functions, can be readily evaluated using only the N_{PCE} deterministic outputs resulting from N_{PCE} original simulations.

3.3.4. Karhunen–Loève expansion for correlated random variables

In the PCE method, ξ must be statistically independent; therefore, correlation should be eliminated when random input variables are correlated. The correlation can be removed using Karhunen–Loève (KL) expansion (Ghanem and Spanos, 1991), also known as principal component analysis or proper orthogonal decomposition. In this study, the correlated uncertain parameters $\boldsymbol{X}$ in the statistical aircraft performance model are linearly transformed into independent random variables ξ by applying the KL expansion:

$$\begin{bmatrix} \xi^{(1)\top} \\ \xi^{(2)\top} \\ \vdots \\ \xi^{(N_{\mathrm{apm}})\top} \end{bmatrix} = \begin{bmatrix} \boldsymbol{X}^{(1)\top} \\ \boldsymbol{X}^{(2)\top} \\ \vdots \\ \boldsymbol{X}^{(N_{\mathrm{apm}})\top} \end{bmatrix} \boldsymbol{W} = \mathcal{X}\boldsymbol{W} \tag{34}$$

where $\mathcal{X}$ is the $N_{\mathrm{apm}} \times N_{\xi}$ raw data matrix, and $\boldsymbol{W}$ is an $N_{\xi} \times N_{\xi}$ square matrix, in which each column vector is an eigenvector of $\mathcal{X}^{\top}\mathcal{X}$. As described in Eq. (34), each sample point of the correlated uncertain parameters within the statistical model of aircraft performance $\boldsymbol{X}^{(i)}$ is linearly transformed into a sample point of the independent random variables $\xi^{(i)}$ $(i = 1, 2, \ldots, N_{\mathrm{apm}})$.

4. β-Hill climbing for trajectory optimization

To optimize the aircraft trajectory, we consider the trajectory optimization problem as follows: determine the decision variables $\boldsymbol{x} = (x_1, x_2, \ldots, x_{N_x})$ that minimize the cost function $J(\boldsymbol{x})$ subject to the boundary conditions $\boldsymbol{x}_{\mathrm{lb}} \le \boldsymbol{x} \le \boldsymbol{x}_{\mathrm{ub}}$, where N_x denotes the number of decision variables, and $\boldsymbol{x}_{\mathrm{lb}}$ and $\boldsymbol{x}_{\mathrm{ub}}$ represent the lower and upper bounds for the decision variables, respectively. This study introduces a robust trajectory optimization approach grounded in metaheuristics. Differing from the population-based algorithms introduced in prior research (Patrón and Botez, 2015; Murrieta-Mendoza et al., 2017, 2018) mentioned in Section 1, a recently introduced sample-based stochastic local search technique known as β-hill climbing (Al-Betar, 2017; Al-Betar et al., 2019) is applied for trajectory optimization. β-hill climbing outperforms other well-known metaheuristic methods such as simulated annealing, genetic algorithms, and particle swarm optimization in searching for optimal solutions (Al-Betar et al., 2019). β-hill climbing excels in traversing extensive search spaces to locate global optima. It leverages the β operator to explore the promising search spaces through a stochastic search, determining whether to replace the current decision variable values with random ones from their feasible regions. In contrast, the $\mathcal{N}$ operator, representing neighborhood search, searches for a neighboring solution that surpasses the current one. The β and $\mathcal{N}$ operators serve as means for exploration and exploitation, respectively, facilitating the efficient search of the optimal solution in β-hill climbing.

The pseudocode for β-hill climbing is presented in Algorithm 1. This process begins with a random solution $\boldsymbol{x}$. Subsequently, a new solution is iteratively produced using the $\mathcal{N}$ and β operators. During the $\mathcal{N}$ operator phase, a random neighboring solution of the current solution $\boldsymbol{x}$ is chosen in the following manner:

$$x_i' = x_i \pm \mathcal{N}_i, \ \exists! i \in [1, N_x] \tag{35}$$

where i is randomly selected from 1 to N_x. The parameter $\mathcal{N}_i$ represents the bandwidth between the current and neighboring solutions for the ith decision variable. Whether to add or subtract $\mathcal{N}_i$ in Eq. (35) is determined randomly. Subsequently, during the β operator phase, a new solution is randomly substituted based on the value of β_k:

$$x_i'' = \begin{cases} x_{\mathrm{lb},i} + r_i(x_{\mathrm{ub},i} - x_{\mathrm{lb},i}) & \text{if } r_i \le \beta_k \\ x_i' & \text{otherwise,} \end{cases} \quad \forall i \in [1, N_x]$$

In every iteration, r_i is produced using a uniformly distributed random number between 0 and 1, denoted as $U(0, 1)$, and β_k represents the β value at the kth iteration. The β value is adjusted deterministically within a predefined range from $\beta_{\min}$ to $\beta_{\max}$:

$$\beta_k = \beta_{\min} + \frac{k}{K}(\beta_{\max} - \beta_{\min})$$

Algorithm 1 Pseudocode for β-hill climbing

Input: Initial decision variables $\boldsymbol{x}$
Output: Optimal decision variables $\boldsymbol{x}$

```
Set β_min, β_max, and K
for i = 1 to N_x do
    r_i = U(0, 1)
    x_i = x_lb,i + r_i(x_ub,i − x_lb,i)
end for
Calculate J(x)
k = 0
while k ≤ K do
    x′ = x
    Randomly choose one decision variable x_i, ∃!i ∈ [1, N_x]
    x′_i = x_i ± N_i
    x″ = x′
    β_k = β_min + (k/K)(β_max − β_min)
    for i = 1 to N_x do
        r_i = U(0, 1)
        if r_i ≤ β_k then
            x″_i = x_lb,i + r_i(x_ub,i − x_lb,i)
        end if
    end for
    Calculate J(x″)
    if J(x″) < J(x) then
        x = x″
        J(x) = J(x″)
    end if
    k = k + 1
end whilereturn x
```

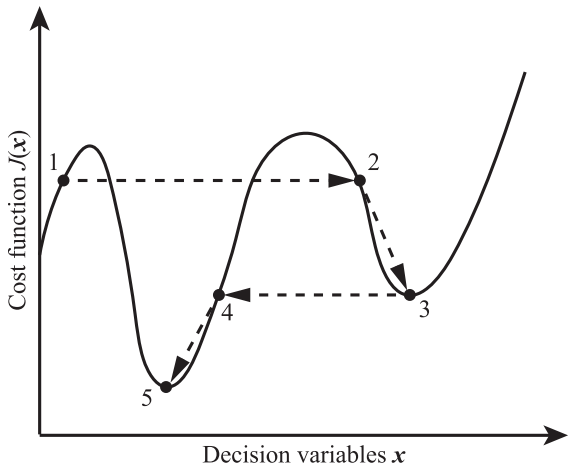

Fig. 4. Schematic diagram of search space navigation using β-hill climbing.

where $\beta_{\min}$ and $\beta_{\max}$ represent the minimum and maximum β values, correspondingly, and K denotes the maximum iteration count. In this study, $\beta_{\min}$ and $\beta_{\max}$ are set to 0.1 and 0.9, respectively, referring to an existing study (Al-Betar et al., 2019). K is determined by evaluating the convergence performance in Section 5. Then, the newly generated solution $\boldsymbol{x}''$ produced by employing the $\mathcal{N}$ and β operators undergoes evaluation through the cost function of the new solution $J(\boldsymbol{x}'')$. If the cost function of the new solution $J(\boldsymbol{x}'')$ surpasses that of the current solution $J(\boldsymbol{x})$, then the current solution $\boldsymbol{x}$ gets substituted with the new solution $\boldsymbol{x}''$. Otherwise, the current solution $\boldsymbol{x}$ remains unchanged. The solution's refinement continues through iteration within the $\mathcal{N}$ operator and/or the β operator until the maximum iteration count is reached.

Fig. 4 provides a simple schematic illustration of search space navigation using β-hill climbing. In this example, the algorithm navigates from solution 1 to solution 5 (global minimum). The β operator helps escape local minima by randomly selecting new

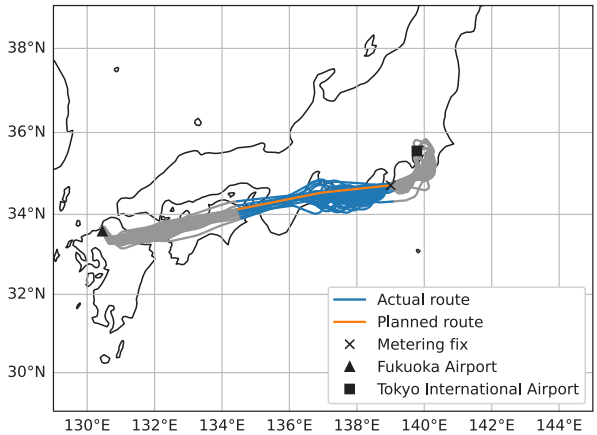

Fig. 5. Planned and actual past routes for 134 flights.

values for decision variables within their feasible regions (e.g., moving from solution 1 to 2, or from 3 to 4). The $\mathcal{N}$ operator, on the other hand, searches for a neighboring solution that improves on the current one (e.g., moving from solution 2 to 3, or from 4 to 5). Thus, these operators facilitate an efficient search for the optimal solution during the β-hill climbing process.

In the robust trajectory optimization problem considered in this study, the cost function $J(\boldsymbol{x})$ is given by

$$J(\boldsymbol{x}) = E\left[\int_0^{t_f} FFdt\right] + \left(E[t_f] - t_{\mathrm{RTA}}\right)^2 \tag{36}$$

where t_f is the terminal time, and t_{RTA} is the RTA at a metering fix, representing the target flight time. $E[t_f]$ and $E\left[\int_0^{t_f} FFdt\right]$ represent the expected values of terminal time and fuel consumption, respectively, across the uncertainty samples in weather prediction and the aircraft performance model used in the simulations. The cost function in Eq. (36) indicates that fuel consumption is minimized while the flight time is kept close to the target time. There are four decision variables, $\boldsymbol{x} = (M_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}})$, and seven decision variables, $\boldsymbol{x} = (M_{\mathrm{crz}}, r_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}}, M_{\mathrm{level}}, h_{\mathrm{level}})$, for continuous and step descent, respectively. The trajectory (s, h_p, v, m) can be calculated using Eqs. (1), (4)–(7), and (9), with consideration to specific weather prediction and aircraft performance parameters, as outlined in Section 2. Eq. (36) is then calculated as the expected value of $N_{\mathrm{ens}} \times N_{\mathrm{PCE}}$ different trajectory calculation results using the ensemble forecasts and the statistical model of aircraft performance by applying aPC as described in Section 3.

5. Numerical simulations

5.1. Problem description

We analyze a popular flight route in Japan, from Fukuoka Airport to Tokyo International Airport (International Air Transport Association, 2019). Fig. 5 illustrates the lateral paths of the planned and actual past routes for the 134 flights considered in this study. The actual flights under current ATM operations do not fly exactly on the planned route, and the flight distance is extended compared to that of the planned route owing to radar vectoring. However, the aircraft is assumed to fly along the planned route using speed control and step descent without extending the flight distance under time-based metering operations. Thus, we investigate the potential fuel savings achieved by adhering to the planned route using speed control and step descent, as opposed to relying on radar vectoring.

Flight data from the on-board data recorders of actual past flights were used. The weight, pressure altitude, and lateral position used in this study were recorded by the flight data recorders at a sampling rate of 1 Hz. The flight data for 134 flights from August 2019 through March 2020 were analyzed in this study. The planned route was obtained from the Japan Aeronautical Information Service Center.[3] The initial and terminal positions on the planned route under metering operations are the existing fixes called YANKS (initial fix) and SPENS (metering fix). The total flight distance is 228.7 NM. We extracted the positions on the actual route that were the closest to the initial and metering fixes, as represented by the colored section of the line in Fig. 5. The initial and terminal altitudes, h_0 and h_f, were set to the altitudes at the positions nearest to the initial and metering fixes obtained from the actual flight data. Note that the initial altitude h_0 is equal to the cruise altitude h_{crz}. The initial weight was also set to the weight at the position closest to the initial fix recorded in the actual flight data. The fuel consumption and flight time for actual past flights were taken from the segment between the nearest positions to the initial and metering fixes.

[3] https://aisjapan.mlit.go.jp

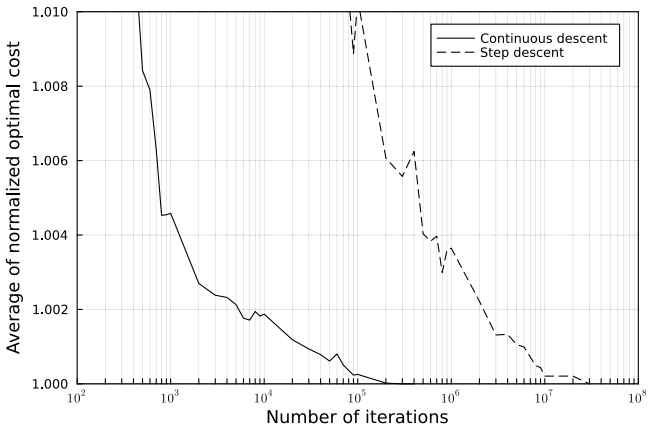

Fig. 6. Convergence behavior (mean of normalized optimal cost).

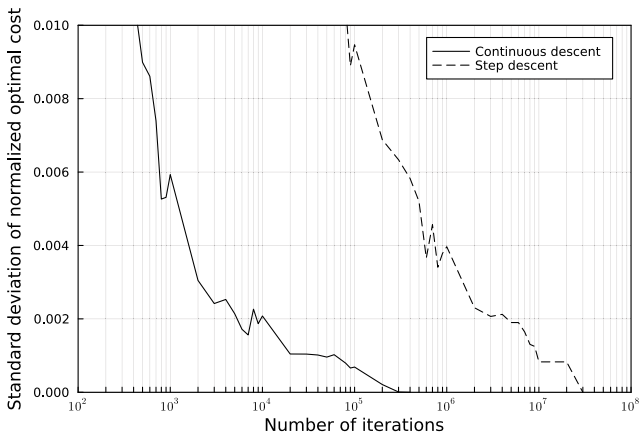

Fig. 7. Convergence behavior (standard deviation of normalized optimal cost).

The robust trajectory optimization was conducted to ensure that the expected value of flight time ($E[t_f]$) was close to the actual flight time obtained from the flight data (t_{RTA}), as shown in Eq. (36). The aircraft type was a Boeing 767-300ER. The upper and lower bounds of h_{level} were h_0 and h_f, respectively. The upper bounds of the CAS and Mach number were set to the maximum operating limit speed of 360 kt and 0.86, respectively (Jenkinson et al., 1999). The lower bounds of the CAS and Mach number were assigned to a speed called the minimum clean or green dot speed, which is given by the speed at which the lift-to-drag ratio is maximized. First, trajectory optimization for continuous descent was performed. Then, if the flight time of the optimal trajectory for continuous descent was not equivalent to the actual flight time ($|E[t_f] - t_{\mathrm{RTA}}| > 5$ s), trajectory optimization for step descent was performed. Step descent can extend the flight time compared to continuous descent.

For aPC, the number of uncertain parameters N_ξ was 12, as mentioned in Section 3.2. In addition, d was set to two, and 91 ($N_{\mathrm{PCE}} = \binom{N_\xi+d}{N_\xi}$) collocation points were derived from the original distribution of the statistical model of aircraft performance represented by 4,000 sample points. This method enabled the assessment of uncertainty propagation in aircraft dynamics. Consequently, instead of conducting 4,000 trajectory calculations, 91 different trajectory calculations were performed to compute the trajectory's expected value. In addition, with respect to the uncertainty in weather prediction, 21 ensemble members were used. Therefore, this study quantified uncertainty propagation and calculated the expected value of the trajectory by conducting 1,911 ($= 21 \times 91$) separate trajectory calculations, combining the uncertainties in weather prediction and aircraft performance model.

The numerical simulations were implemented in Julia and conducted on an Intel Core i7-11800H processor with 64 GB of memory.

5.2. Results and discussion

We evaluated the convergence performance of the β-hill climbing algorithm used for the trajectory optimization. Figs. 6 and 7 show the convergence behaviors of the trajectory optimization for two specific flights, one using continuous descent and one using step descent. The number of iterations K in the β-hill climbing algorithm varied from 10^2 to 10^8, and 100 Monte Carlo simulations

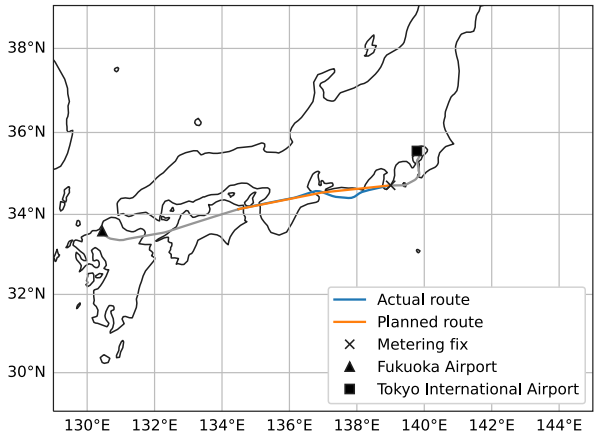

Fig. 8. Actual route (average case).

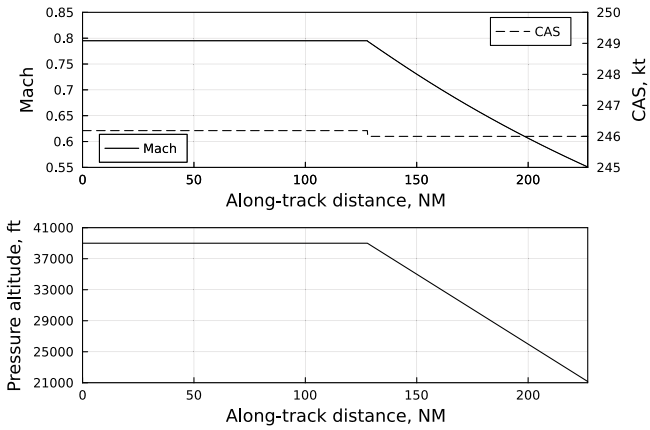

Fig. 9. Optimal trajectory (average case).

of trajectory optimization were performed for each iteration. The optimal cost at each K is normalized using the mean optimal cost for the case of $K = 10^8$, and Figs. 6 and 7 show the mean and standard deviation, respectively, of the normalized optimal costs over 100 Monte Carlo simulations. The solid and dashed lines show the results of the trajectory optimization for continuous and step descent, respectively. As K increases, the mean approaches the same value and, accordingly, the standard deviation decreases, as shown in Figs. 6 and 7. Figs. 6 and 7 indicate that convergence was achieved at $K = 4 \cdot 10^5$ for continuous descent and at $K = 3 \cdot 10^7$ for step descent. In the following simulations, these values of K were used for trajectory optimization.

We evaluated the 134 actual past flights with path stretching, as shown in Fig. 5. For the numerical simulations, we extracted the positions on the actual route that were closest to the initial and metering fixes, as represented by the colored section of the line in Fig. 5. The flight time and fuel consumption of the extracted segment of the actual route were compared with those of the optimal trajectory. We generated optimal trajectories that achieved the same flight times as the actual past flights through continuous descent on the planned route without path stretching. Below are examples of the average and maximum cases of additional distance among the 134 flights. The additional distance refers to the path stretching distance, calculated as the difference between the distance of the planned route and that of the actual flight.

Fig. 8 illustrates the actual route for the average case of additional distance. Here, the additional distance is 6.86 NM. The trajectory optimization results for the average case are presented in Fig. 9. The optimal decision variables were $(M_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}}) = (0.795, 1.7, 0.794, 246)$. Although the actual past route is extended compared to the planned route, the same flight time can be achieved using continuous descent alone, that is, speed control on the planned route without path stretching. However, in some cases, continuous descent did not achieve the same flight time as the actual past flights.

Next, Fig. 10 illustrates the actual route for the maximum additional distance case. The actual past route in the maximum case was more extended than the actual route in the average case. Here, the additional distance is 40.74 NM. Fig. 11 presents the trajectory optimization results for the maximum case. The optimal decision variables were $(M_{\mathrm{crz}}, r_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}}, M_{\mathrm{level}}, h_{\mathrm{level}}) = (0.709, 0.27, 2.2, 0.707, 227, 0.586, 29,000)$. In Fig. 11, the aircraft accelerates from $M_{\mathrm{level}} = 0.586$ (approximately 222 kt CAS) during level flight at $h_{\mathrm{level}} = 29{,}000$ ft to $CAS_{\mathrm{des}} = 227$ kt in the subsequent descent. This acceleration is a result of the constraints applied

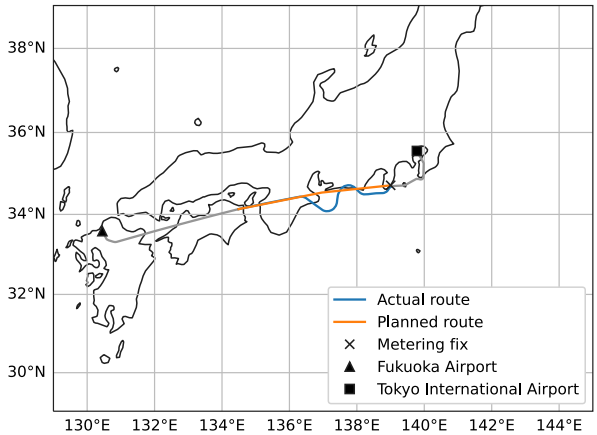

Fig. 10. Actual route (maximum case).

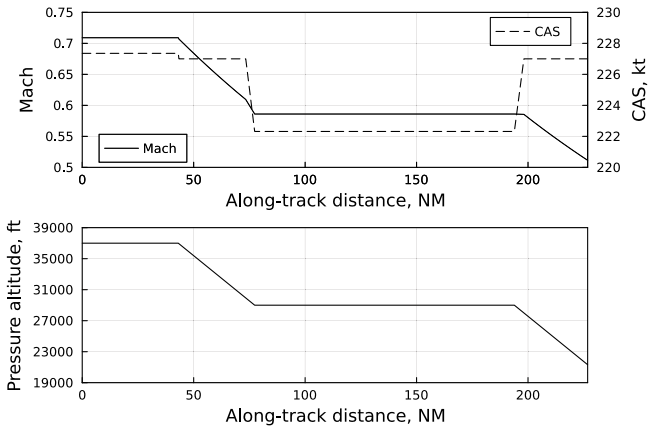

Fig. 11. Optimal trajectory (maximum case).

during trajectory optimization, where both CAS_{des} and γ_{fix} were assumed to remain constant before and after the level-off, as illustrated in Fig. 1. The descent segment following the level-off continues with the same optimized $CAS_{\mathrm{des}} = 227$ kt, which was determined by including the descent phase before the level-off. It is important to note that allowing for additional flexibility – such as individualizing the CAS and fixed-FPA before and after the level-off – could potentially yield an alternative optimal solution that does not involve acceleration after the level-off. However, under the current set of constraints, the solution shown in Fig. 11 is the optimal trajectory. In the maximum case, because the same flight time cannot be achieved solely through speed control, it is accomplished using step descent, which allows for a longer flight time. The same flight time could not be achieved through continuous descent in four of the 134 flights, and step descent was required to extend the flight time.

Fig. 12 presents the fuel consumption of the optimal solution for the average case. The histogram indicates the variation due to the uncertainties in weather prediction and aircraft performance model, and illustrates all $84,000 (= 21 \times 4,000)$ results from the combination of 21 ensemble forecasts and 4,000 sample points in the aircraft performance model. The solid line represents the expected value approximated by the aPC algorithm (approximated value), and the dashed line represents the exact mean of all 84,000 results (exact mean). Note that the solid and dashed lines in Fig. 12 overlap. Fig. 12 clearly shows that aPC provides a fairly good approximation of the expected fuel consumption value. In addition, we compared the exact means and approximated values of fuel consumption with data from the 134 flights. The root mean squared error between the exact means and approximated values is $4.52 \cdot 10^{-2}$ kg, and the root mean squared percentage error is $3.41 \cdot 10^{-5}$, which is sufficiently small to suggest that the aPC algorithm can accurately approximate the fuel consumption. The aPC algorithm requires only 91 collocation points to obtain a good approximate solution; it does not need to use all 4,000 sample points of the aircraft performance model. Using all 4,000 sample points is computationally expensive and intractable, especially in an iterative process such as trajectory optimization. In terms of computational time for obtaining each optimal trajectory, applying aPC took 55.1 s for continuous descent (average case, Figs. 8 and 9) and 1065.1 s for step descent (maximum case, Figs. 10 and 11). Trajectory optimization for step descent requires more time due to the increased number of decision variables and iterations needed for convergence. In contrast, using all 4,000 sample points required 2218.7 s for continuous descent and 42404.8 s for step descent. These computational times are the averages of 10 identical

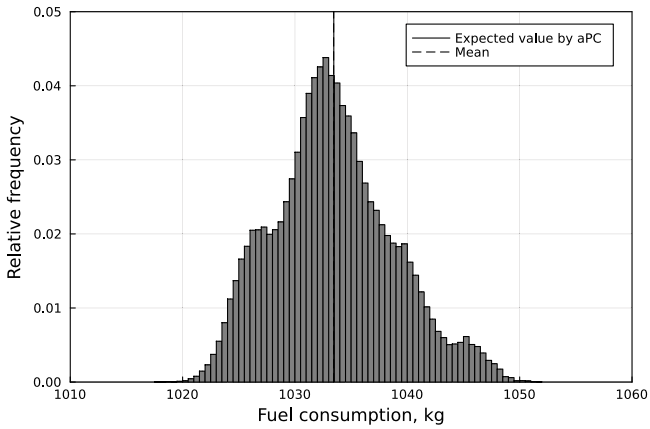

Fig. 12. Fuel consumption of optimal trajectory and expected value approximated by aPC algorithm (average case).

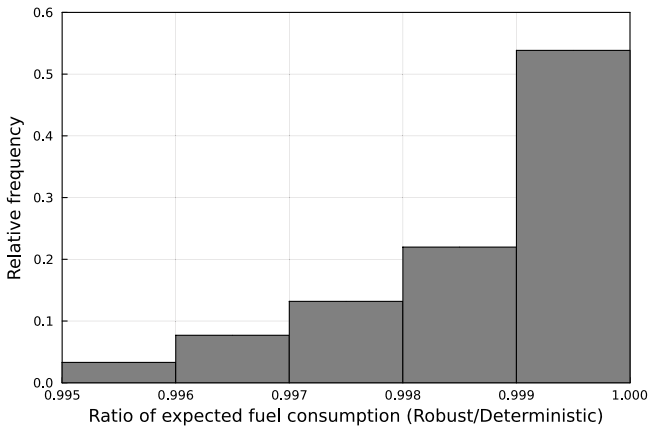

Fig. 13. Histogram of the ratio of expected fuel consumption (robust vs. deterministic) for 134 flights.

optimization runs. Thus, applying aPC reduces computational time by approximately 98%. Therefore, our proposed algorithm based on the aPC method can provide accurate approximate solutions while significantly improving computational efficiency.

To further validate the proposed robust trajectory optimization, we compared it with deterministic trajectory optimization. For the deterministic model, we used the average weather forecasts (the mean of 21 ensemble members) and the average aircraft performance model (the mean of all 4,000 sample points). The cost function in Eq. (36) becomes deterministic, simplifying $J(\boldsymbol{x}) = \int_0^{t_f} FFdt + \left(t_f - t_{\mathrm{RTA}}\right)^2$. The optimal decision variables obtained through deterministic trajectory optimization for the average and maximum cases are $(M_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}}) = (0.798, 1.7, 0.791, 245)$ and $(M_{\mathrm{crz}}, r_{\mathrm{crz}}, \gamma_{\mathrm{fix}}, M_{\mathrm{des}}, CAS_{\mathrm{des}}, M_{\mathrm{level}}, h_{\mathrm{level}}) = (0.71, 0.26, 2.2, 0.71, 228, 0.586, 29,000)$, respectively. These results show that the solutions from deterministic optimization slightly differ from those obtained through robust optimization. Next, we compared fuel consumption. For the deterministic optimal solution, we evaluated the effect of uncertainties in weather prediction and the aircraft performance model by conducting a posterior evaluation using the full set of uncertainties (21 ensemble forecasts combined with 4,000 sample points from the aircraft performance model, yielding 84,000 results) to calculate the expected value of fuel consumption (the average of all 84,000 results). Fig. 13 presents a histogram of the ratio of expected fuel consumption from robust trajectory optimization relative to that from deterministic trajectory optimization for the 134 flights. If the ratio is less than 1, robust trajectory optimization results in lower expected fuel consumption than the deterministic approach. The average ratio across the 134 flights was 0.998, with a minimum of 0.995. This indicates that the proposed robust trajectory optimization, by incorporating uncertainties, provides an optimal trajectory that is expected to save more fuel – on average, about 0.2%, with a maximum saving of about 0.5% – compared to the deterministic approach.

We then evaluated the fuel savings for the aforementioned 134 flights. Continuous or step descent reduces fuel consumption by eliminating path stretching. Fig. 14 shows scatter plots of the additional distance and fuel savings for individual flights. The fuel consumption of the optimal solution was compared with the actual fuel consumption recorded in the flight data. In addition, note that fuel savings are defined as the reduction in fuel consumption for the segment of metering operations relative to the gate-to-gate fuel consumption. The expected value and standard deviation of fuel savings for each flight can be determined using the variation

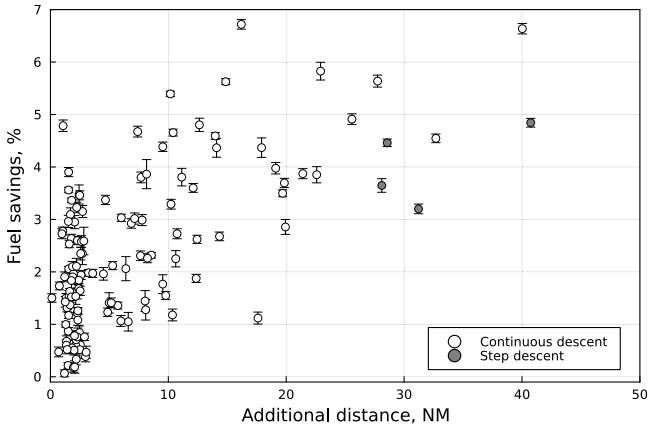

Fig. 14. Fuel savings for 134 flights.

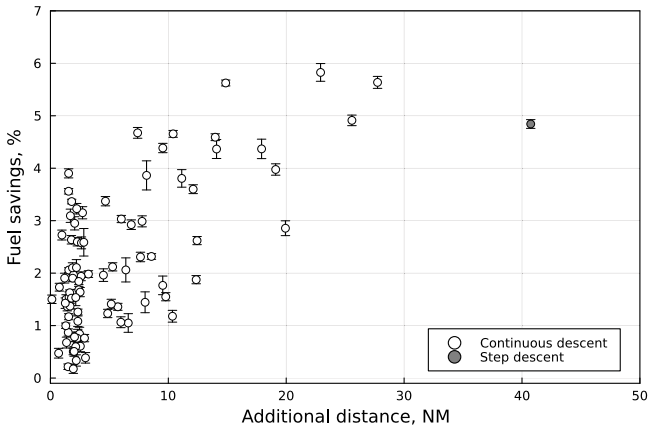

Fig. 15. Fuel savings for 91 flights with path stretching and continuous descent.

in fuel consumption due to the uncertainties in weather forecasts and aircraft performance model demonstrated in Fig. 12. Fig. 14 illustrates that the dots denote the expected fuel savings, while the error bars depict the standard deviations. As the impact of weather uncertainty is different for each flight, the standard deviation of fuel savings also varies from flight to flight. For the flights analyzed in this study, continuous descent was feasible for the majority of flights, particularly those with an additional distance of approximately 25 NM or less. Even for the flights with an additional distance exceeding 30 NM, continuous descent alone was feasible in some cases, depending on the wind conditions and cruise altitude. There was a tendency for greater fuel savings as the additional distance increased. The average fuel savings were $2.34 \pm 0.11\%$. Here, the average fuel savings were calculated from the average of the expected values and variances of fuel savings for each flight.

Among the 134 historical flight data sets, some involve not only path stretching but also step descent. To ensure a fairer comparison, we evaluated the 91 flights involving path stretching combined with continuous descent only. Fig. 15 shows the fuel savings from the optimal trajectories for these 91 flights, with an average savings of $2.19 \pm 0.10\%$, which is slightly lower than the savings for all 134 flights. This suggests that the fuel savings from trajectory optimization for the past flights with step descent were significant. While the fuel savings are slightly smaller when focusing on flights with path stretching and continuous descent, the results still demonstrate that savings can be achieved through speed control on the originally planned route instead of path stretching. Although path stretching is commonly used in current ATM operations for flight time management, additional fuel is required for path stretching. When imposing the same amount of delay, speed control on the planned route is expected to reduce fuel consumption. It is clear that applying speed control and step descent is effective for fuel savings. Thus, these numerical simulations allowed us to assess the fuel savings and feasibility of speed control and step descent.

Finally, with regard to computational time for trajectory optimization, as mentioned earlier, the optimal trajectory for continuous descent can be obtained in about 55 s, which may be an acceptable duration for dynamically applying the trajectory optimization using aPC in actual operations. However, the optimal trajectory for step descent takes about 1065 s, presenting challenges for real-time application of the proposed method. Future studies will explore leveraging GPUs to accelerate the process, as demonstrated by González-Arribas et al. (2023). Additionally, this study focused on evaluating the potential fuel savings by optimizing the vertical

path along the initially planned route without path stretching. However, as discussed in Section 1, the study by Sáez and Prats (2023) suggests that path stretching, when combined with idle thrust descents along renegotiated, longer routes, does not necessarily increase fuel consumption. Even with path stretching, some flights evaluated in this study may still further reduce fuel consumption. Further analysis in future studies is needed to derive the optimal strategy, including consideration of the lateral path. Furthermore, it should be noted that the numerical simulations conducted in this study showed the maximum potential for fuel savings. This study primarily concentrated on assessing the fuel savings for individual flights through continuous and step descent without path stretching in the presence of uncertainties in weather prediction and aircraft performance models. Fuel savings were analyzed for 134 flights along the same route. However, because the evaluation was conducted solely for individual flights, the potential conflict between aircraft resulting from continuous and step descent was not considered. In practical operations, ensuring aircraft separation is crucial before changing the speed and/or altitude for time-based metering. Therefore, future studies will address potential conflicts in implementing speed control and step descent for time-based metering in real-world operations. A traffic simulation will be conducted to investigate this issue quantitatively.

6. Conclusions

This paper proposed a robust trajectory optimization algorithm for optimizing flight profiles under arbitrary uncertainties in the weather forecasting and aircraft performance models. Ensemble forecast data were used to provide the uncertainty in the weather forecasts, and a statistical model of aircraft performance estimated from flight data was used to provide the uncertainty in the aircraft performance model. The ensemble forecast data consists of 21 members; we can obtain 21 different outcomes from trajectory calculations and utilize them to determine the expected value of the trajectory under weather prediction uncertainty. However, it is computationally burdensome to calculate results for every sample point in the arbitrary probability distributions for the statistical aircraft performance model. The proposed method incorporates aPC, which can handle arbitrary uncertainties. aPC was employed to efficiently quantify arbitrary uncertainties in aircraft dynamics. The β-hill climbing algorithm was employed to determine the optimal aircraft trajectory, leveraging its efficient exploration of extensive search spaces to determine the optimal solution.

Using actual past flight data, we conducted numerical simulations of popular domestic flights in Japan to evaluate the potential fuel savings achieved by speed control and step descent on the planned route instead of using path stretching in time-based metering operations. aPC provided a fairly good approximation of the expected fuel consumption. The 134 past flights with path stretching were evaluated, yielding an average fuel savings of $2.34 \pm 0.11\%$. Some of these past flights also included step descent. Therefore, we focused on the 91 flights with only path stretching and continuous descent, which had an average fuel savings of $2.19 \pm 0.10\%$. Although path stretching is commonly used in current ATM operations for flight time management, additional fuel is required for this approach. When imposing the same amount of delay, speed control and step descent on the planned route are expected to improve fuel consumption. Thus, the speed control and step descent strategies have been shown to be effective in improving fuel savings. Moreover, fuel savings could also incentivize airlines to implement time-based metering operations. The results of this study can provide valuable guidance for implementing metering operations.

In this study, we focused on a flight-by-flight evaluation of fuel savings through speed control and step descent in the presence of uncertainties in the weather prediction and aircraft performance model. However, because each flight's assessment was conducted independently, the possibility of conflict between aircraft resulting from speed control and step descent was not considered. Therefore, our future studies will address the potential conflict with implementing speed control and step descent for time-based metering in actual operational scenarios.

CRediT authorship contribution statement

Yoshinori Matsuno: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Conceptualization. **Haruki Matsuda:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Nomenclature

In this appendix, Table A.1 summarizes the nomenclature used in the trajectory calculation.

Table A.1
Nomenclature used in trajectory calculation.

Symbols	Definitions
CAS_{des}	descent calibrated airspeed
D	drag force
FF	fuel flow
g	acceleration due to gravity
h	geodetic altitude
h_{crz}	cruise altitude
h_f	terminal altitude at a metering fix
h_{level}	level-off altitude
h_p	pressure altitude
m	aircraft weight
M_{crz}	cruise Mach number
M_{des}	descent Mach number
M_{level}	Mach number during level-off
r_{crz}	ratio of cruise distance to cruise and level-off distances
$ROCD$	rate of climb or descent
s	along-track distance
s_{crz}	cruise distance
s_f	along-track distance to a metering fix
t	time
t_f	terminal time at a metering fix
t_{RTA}	required time of arrival at a metering fix
T	temperature in a non-standard atmosphere
T_{ISA}	temperature in the International Standard Atmosphere
Th	thrust force
v	true airspeed
v_g	ground speed
w_a	along-track wind
w_c	cross-track wind
w_e	eastward wind
w_n	northward wind
γ_a	air flight path angle
γ_{fix}	fixed flight path angle (FPA) in fixed-FPA descents
χ	heading angle
ψ	true track angle

Data availability

The authors do not have permission to share data.

References

Al-Betar, M.A., 2017. β-Hill climbing: An exploratory local search. Neural Comput. Appl. 28, 153–168.

Al-Betar, M.A., Aljarah, I., Awadallah, M.A., Faris, H., Mirjalili, S., 2019. Adaptive β-hill climbing for optimization. Soft Comput. 23, 13489–13512.

Bauer, P., Thorpe, A., Brunet, G., 2015. The quiet revolution of numerical weather prediction. Nature 525 (7567), 47–55.

Cerezo-Magaña, M., Olivares, A., Staffetti, E., 2022. A stochastic switched optimal control approach to formation mission design for commercial aircraft. IEEE Trans. Aerosp. Electron. Syst. 58 (5), 4342–4360.

Clarke, J.-P.B., Ho, N.T., Ren, L., Brown, J.A., Elmer, K.R., Tong, K.-O., Wat, J.K., 2004. Continuous descent approach: Design and flight test for Louisville International Airport. J. Aircr. 41 (5), 1054–1066.

Cook, A., Blom, H.A.P., Lillo, F., Mantegna, R.N., Micciche, S., Rivas, D., Vázquez, R., Zanin, M., 2015. Applying complexity science to air traffic management. J. Air Transp. Manag. 42, 149–158.

Dahlberg, J., Granberg, T.A., Polishchuk, T., Schmidt, C., Sedov, L., 2018. Capacity-driven automatic design of dynamic aircraft arrival routes. In: 2018 IEEE/AIAA 37th Digital Avionics Systems Conference. DASC, pp. 1–9.

Dalmau, R., Melgosa, M., Vilardaga, S., Prats, X., 2018. A fast and flexible aircraft trajectory predictor and optimiser for ATM research applications. In: Proceedings of the 8th International Conference on Research in Air Transportation. Barcelona, Spain, pp. 1–8.

Dalmau, R., Prats, X., 2017. Controlled time of arrival windows for already initiated energy-neutral continuous descent operations. Transp. Res. C 85, 334–347.

Dalmau, R., Prats, X., Ramonjoan, A., Soley, S., 2020. Estimating fuel consumption from radar tracks: a validation exercise using FDR and radar tracks from descent trajectories. CEAS Aeronaut. J. 11 (2), 355–365.

De Smedt, D., Bronsvoort, J., McDonald, G., 2013. Controlled time of arrival feasibility analysis. In: Proceedings of the Tenth USA/Europe Air Traffic Management Research and Development Seminar. Chicago, IL, USA, pp. 1–10.

del Pozo Domínguez, M., Leonés, J.L., Morales, M.P., Astorga, R.V., 2023. Assessing the environmental impact of continuous descent operations based on quick access recorder and surveillance data. In: Proceedings of the 2023 IEEE/AIAA 42nd Digital Avionics Systems Conference. DASC, pp. 1–9.

Delgado, L., De La Torre, D., Kuljanin, J., Prats, X., 2024. Considering expected TMA holding into in-flight trajectory optimization. Trans. Japan Soc. Aeronaut. Space Sci. 67 (3), 109–118.

Delgado, L., Prats, X., 2012. En route speed reduction concept for absorbing air traffic flow management delays. J. Aircr. 49 (1), 214–224.

EUROCONTROL Experimental Centre, 2010. Revision of atmosphere model in BADA aircraft performance model.

Filippone, A., 2017. Options for aircraft noise reduction on arrival and landing. Aerosp. Sci. Technol. 60, 31–38.

Franco, A., Rivas, D., Valenzuela, A., 2018. Probabilistic aircraft trajectory prediction in cruise flight considering ensemble wind forecasts. Aerosp. Sci. Technol. 82, 350–362.
Gelman, A., Lee, D., Guo, J., 2015. Stan: A probabilistic programming language for Bayesian inference and optimization. J. Educ. Behav. Stat. 40 (5), 530–543.
Ghanem, R.G., Spanos, P.D., 1991. Stochastic Finite Elements: A Spectral Approach. Springer New York, NY.
Ghazi, G., Botez, R.M., Bourrely, C., Turculet, A.-A., 2021. Method for calculating aircraft flight trajectories in presence of winds. J. Aerosp. Inf. Syst. 18 (7), 442–463.
González-Arribas, D., Baneshi, F., Andrés, E., Soler, M., Jardines, A., García-Heras, J., 2023. Fast 4D flight planning under uncertainty through parallel stochastic path simulation. Transp. Res. C 148, 104018.
González-Arribas, D., Soler, M., Sanjurjo-Rivo, M., 2018. Robust aircraft trajectory planning under wind uncertainty using optimal control. J. Guid. Control Dyn. 41 (3), 673–688.
Gudmundsson, S.V., Cattaneo, M., Redondi, R., 2021. Forecasting temporal world recovery in air transport markets in the presence of large economic shocks: The case of COVID-19. J. Air Transp. Manag. 91, 102007.
Harada, A., Miyamoto, Y., Miyazawa, Y., Funabiki, K., 2013. Accuracy evaluation of an aircraft performance model with airliner flight data. Trans. Jpn. Soc. Aeronaut. Space Sci., Aerosp. Technol. Jpn. 11, 79–85.
International Air Transport Association, 2019. World air transport statistics 2019.
International Civil Aviation Organization, 2010. Continuous descent operations (CDO) mannual.
Iwata, D., Nonaka, Y., Funai, Y., Shindo, T., Tanaka, S., Sato, M., Itoh, E., 2024. Demonstration of fixed flight-path angle descent via scheduled commercial flights. J. Air Transp. 32 (3), 130–145.
Jenkinson, L.R., Simpkin, P., Rhodes, D., 1999. Civil Jet Aircraft Design. Butterworth-Heinemann.
Kamo, S., Rosenow, J., Fricke, H., Soler, M., 2021. Robust CDO trajectory planning under uncertainties in weather prediction. In: Proceedings of the Fourteenth USA/Europe Air Traffic Management Research and Development Seminar. Virtual Event, pp. 1–10.
Li, H., Zhang, D., 2007. Probabilistic collocation method for flow in porous media: Comparisons with other stochastic methods. Water Resour. Res. 43, 1–13.
Matsuda, H., Matsuno, Y., 2023. Statistical modeling of aircraft performance for airliner operations. Aeronaut. Space Sci. Jpn. 71 (7), 185–189.
Matsuno, Y., Andreeva-Mori, A., 2020. Analysis of achievable airborne delay and compliance rate by speed control: A case study of international arrivals at Tokyo International Airport. IEEE Access 8, 90686–90697.
Matsuno, Y., Andreeva-Mori, A., 2022. Feasibility analysis of time-based metering using cruise-speed control and flight-level change for domestic flights in Japan. Trans. Japan Soc. Aeronaut. Space Sci. 65 (4), 139–146.
Matsuno, Y., Tsuchiya, T., Wei, J., Hwang, I., Matayoshi, N., 2015. Stochastic optimal control for aircraft conflict resolution under wind uncertainty. Aerosp. Sci. Technol. 43, 77–88.
Murrieta-Mendoza, A., Botez, R.M., Bunel, A., 2018. Four-dimensional aircraft en route optimization algorithm using the artificial bee colony. J. Aerosp. Inf. Syst. 15 (6), 307–334.
Murrieta-Mendoza, A., Hamy, A., Botez, R.M., 2017. Four- and three-dimensional aircraft reference trajectory optimization inspired by ant colony optimization. J. Aerosp. Inf. Syst. 14 (11), 597–616.
Nuic, A., Poles, D., Mouillet, V., 2010. BADA: An advanced aircraft performance model for present and future ATM systems. Internat. J. Adapt. Control Signal Process. 24 (10), 850–866.
Oladyshkin, S., Nowak, W., 2012. Data-driven uncertainty quantification using the arbitrary polynomial chaos expansion. Reliab. Eng. Syst. Saf. 106, 179–190.
Park, S.G., Dutta, P., Menon, P.K., 2017. Optimal trajectory option sets for in-flight climb-descend trajectory negotiations. In: Proceedings of the 17th AIAA Aviation Technology, Integration, and Operations Conference. Denver, CO, USA, pp. 1–17.
Patrón, R.S.F., Botez, R.M., 2015. Flight trajectory optimization through genetic algorithms for lateral and vertical integrated navigation. J. Aerosp. Inf. Syst. 12 (8), 533–544.
Prats, X., Hansen, M., 2011. Green delay programs absorbing ATFM delay by flying at minimum fuel speed. In: Proceedings of the Ninth USA/Europe Air Traffic Management Research and Development Seminar. Berlin, Germany, pp. 1–8.
Sáez, R., Polishchuk, T., Schmidt, C., Hardell, H., Smetanová, L., Polishchuk, V., Prats, X., 2021. Automated sequencing and merging with dynamic aircraft arrival routes and speed management for continuous descent operations. Transp. Res. C 132, 103402.
Sáez, R., Prats, X., 2023. Time-based-fuel-efficient aircraft descents: Thrust-idle descents along re-negotiated routes vs. powered descents along published routes. Transp. Res. Part D: Transp. Environ. 114, 103563.
Sáez, R., Prats, X., Polishchuk, T., Polishchuk, V., Schmidt, C., 2020. Automation for separation with continuous descent operations: Dynamic aircraft arrival routes. J. Air Transp. 28 (4), 144–154.
Soler, M., González-Arribas, D., Sanjurjo-Rivo, M., García-Heras, J., Sacher, D., Gelhardt, U., Lang, J., Hauf, T., Simarro, J., 2020. Influence of atmospheric uncertainty, convective indicators, and cost-index on the leveled aircraft trajectory optimization problem. Transp. Res. C 120, 102784.
Toratani, D., Wickramasinghe, N.K., Westphal, J., Feuerle, T., 2020. Feasibility study on applying continuous descent operations in congested airspace with speed control functionality: Fixed flight-path angle descent. Aerosp. Sci. Technol. 107, 106236.
Wiener, N., 1938. The homogeneous chaos. Amer. J. Math. 60 (4), 897–936.
World Meteorological Organization, 2012. Guidelines on ensemble prediction systems and forecasting.
Wu, M.G., Green, S.M., Jones, J., 2015. Strategies for choosing descent flight-path angles for small jets. J. Aircr. 52 (3), 847–866.
Wubben, F., Busink, J., 2000. Environmental Benefits of Continuous Descent Approaches at Schiphol Airport Compared with Conventional Approach Procedures. Technical Report NLR-TP-2000-275.
Xiu, D., Hesthaven, J.S., 2005. High-order collocation methods for differential equations with random inputs. SIAM J. Sci. Comput. 27, 1118–1139.
Xiu, D., Karniadakis, G.E., 2002. The Wiener–Askey polynomial chaos for stochastic differential equations. SIAM J. Sci. Comput. 24 (2), 619–644.
Xu, Y., Dalmau, R., Prats, X., 2017. Maximizing airborne delay at no extra fuel cost by means of linear holding. Transp. Res. C 81, 137–152.